

Cortex-M Online Programmer

User Manual

Rev2.12 July 2023

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1 Introduction

1.1 Overview

CM ISP (Cortex-M In-System Programmer) is an online programmer software for Cortex-M series MCU of Xiaohua Semiconductor (XHSC), supporting all Cortex-M series MCU products of Xiaohua Semiconductor. This article will introduce the usage of the online programmer software (xhsc.exe) and programming precautions.

This article is applicable to the online programmer software version number V2.02 and above.

1.2 Connection Overview

When using CM ISP programmer software (xhsc.exe), the connection between the serial port module and the target MCU is shown in Figure 1-1.

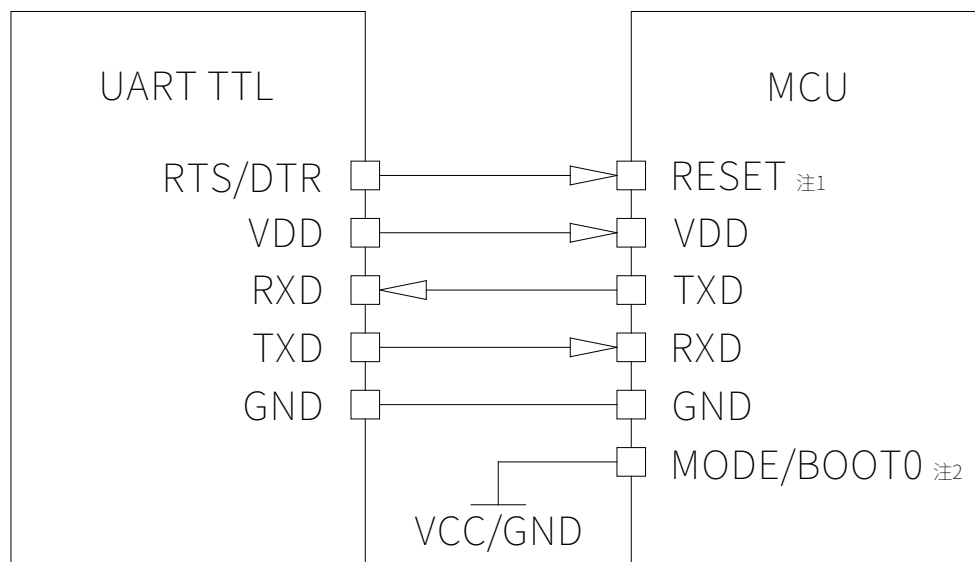


Figure 1-1 Connection between the serial port module and the target MCU

Connect the target chip to the serial port module, and then connect the serial port module to the PC.

Notes:

- Some series do not have MODE/BOOT0 pins. When serial programming, the RTS or DTR of the serial port module needs to be connected to the RESET pin of the target MCU.
- The connection method of the MODE/BOOT0 pin of different chip models may be different. Please refer to the programmer software interface prompt for details.

1.3 Software Operation Overview

The operating environment of the online programmer software is shown in Table 1-1.

Table 1-1 Operating environment of the programmer software

Operating system	Windows 7, Windows 8, Windows 10
Framework version	Framework 4.0 or above

Running the software requires that Microsoft.NET Framework v4.0 or above has been installed on the computer. Please confirm whether Framework 4.0 exists in the computer system path "C:\Windows\Microsoft.NET\Framework(64)", as shown in Figure 1-2.

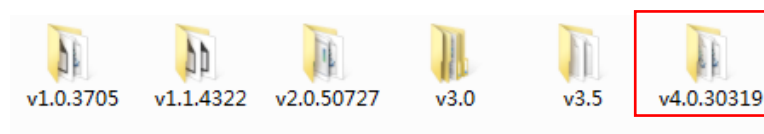


Figure 1-2 Framework 4.0

If the operating system is not installed, please select the corresponding version on the Microsoft official website to download.

The file directory of the folder where the online programmer software is located is shown in Figure 1-3.

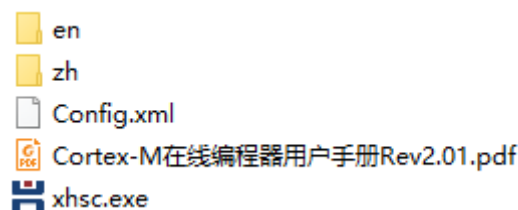


Figure 1-3 Online programmer file directory

The "en" and "zh" folders contain program language configuration files, which can realize the switching between Chinese and English in the program interface, but are not required.

"Config.xml" is the user configuration file. It is automatically generated after the program is closed to save user-related settings. This file is not available for the first use.

"Cortex-M Online Programmer User Manual Rev2.01.pdf" is the user manual file, which can be opened by clicking the "Help" menu in the program interface.

"xhsc.exe" is the program execution file. This file can also be run alone, but it cannot realize the switching between Chinese and English, and only supports English display.

Double-click "xhsc.exe" to open the software. The software interface is shown in Figure 1-4.

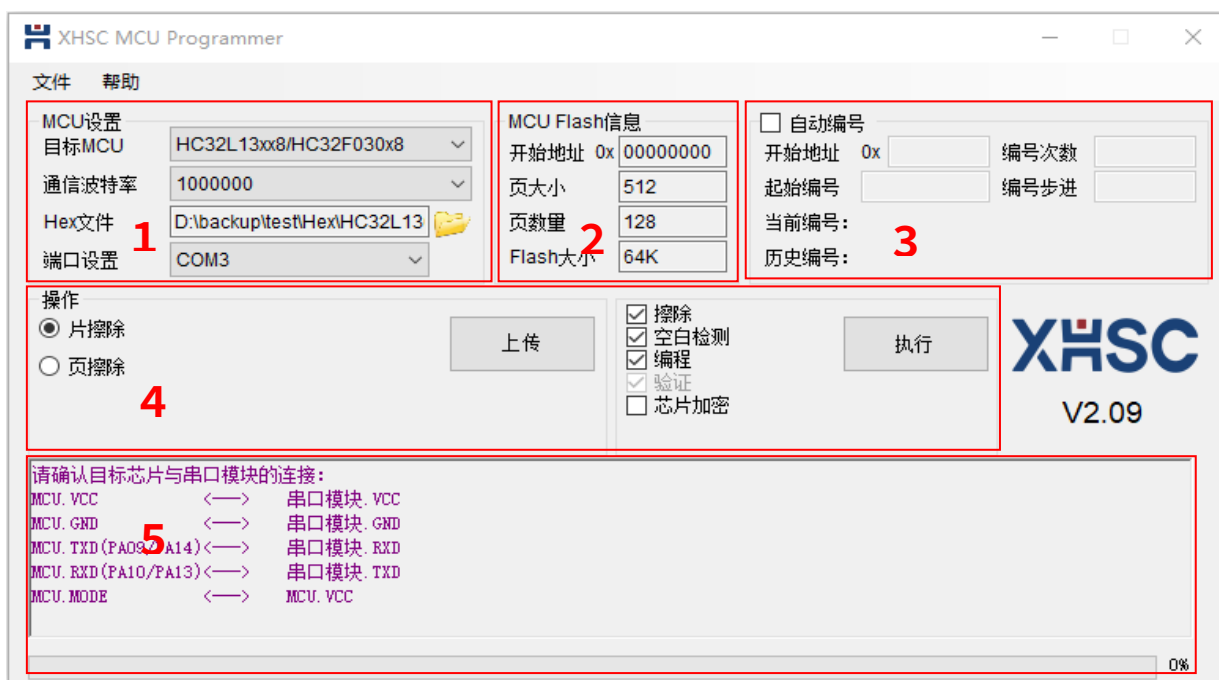


Figure 1-4 Software Interface 1

- **MCU Settings:** used to set the target MCU model, the crystal frequency used by the target MCU (HC32F146xA/ HC32M140xA and HC32F146x8/ HC32M140x8) or the communication baud rate of the selected serial port (other series), the Hex file to be burned, and the COM port number on the PC.
- **MCU Flash Information:** Displays the selected MCU Flash information: start address, page size, number of pages, and Flash size.
- **Auto Numbering:** Used for users to number the MCU.
- **Operation:** The operation is divided into two parts: upload and execution. Upload is to read the target MCU Flash data and save it as a .hex file;
Execution can select the operation item you want to operate, including erase, blank check, programming (verification), and chip encryption. After selecting the operation item, click the execute button.
- **Information Display:** Used to display programming information.

For target MCU HC32F460xExx/HC32F45xxExx, the software interface is shown in Figure 1-5.



Figure 1-5 Software Interface 2

HC32F460xExx/HC32F45xxExx has read protection 1, read protection 2, and encryption functions. If you want to operate these functions, you need to select the corresponding option box. Read protection 1 requires a password, and the encryption function requires setting the flash encryption range.

2 Quick Operation

The following will describe how to quickly program.

1. Connect the serial port of the USB-to-serial cable to the serial programming pin of the target MCU. Taking the HC32L136 series as an example, the connection method is shown in Figure 2-1. Pull up the target MCU MODE pin, and then power on the target MCU to enter the serial programming mode.

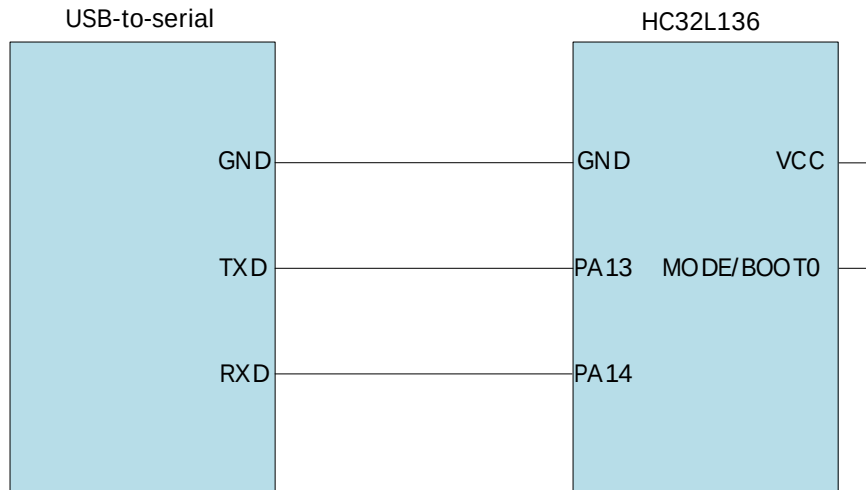


Figure 2-1 Hardware connection

2. Connect the PC and MCU target board with a USB-to-serial cable. Open the software, select the corresponding target MCU model, set the communication baud rate, select the programmed Hex file, and set the port to the COM port number used.



Figure 2-2 MCU settings

3. 3. Select the desired operation. For example, select the "Erase", "Blank Detection", "Program (Verification)" checkboxes. As shown in Figure 2-3.

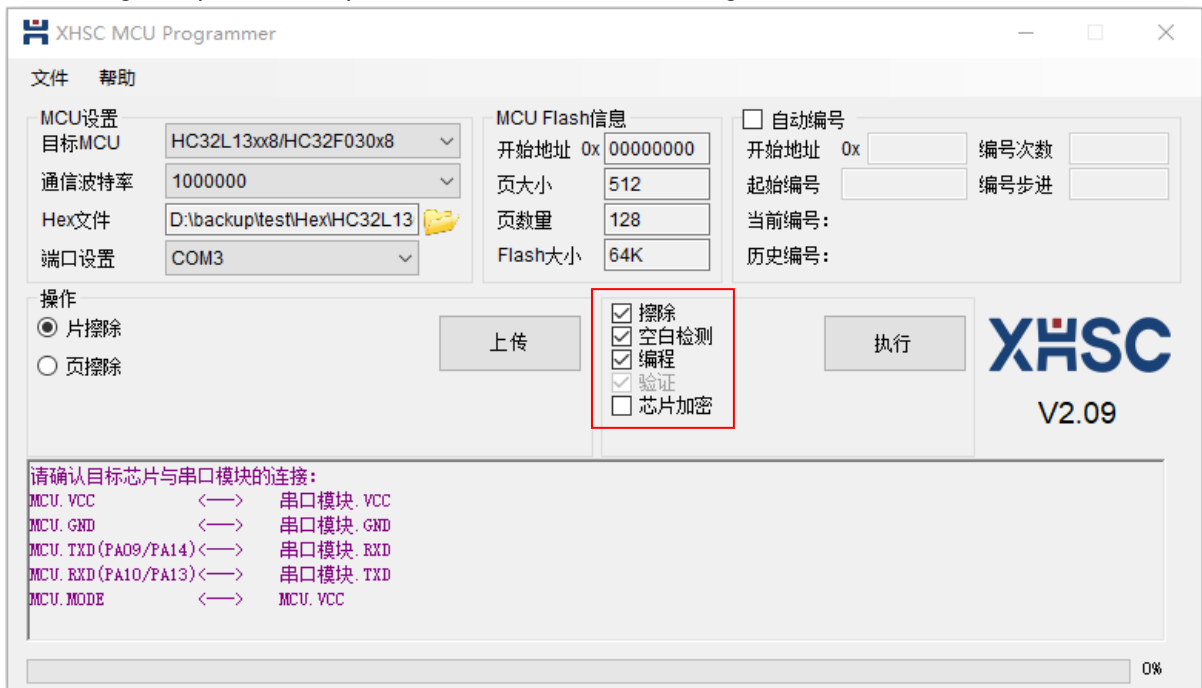


Figure 2-3 Select Operation

4. Click the "Execute" button to start programming and wait for the programming to end.

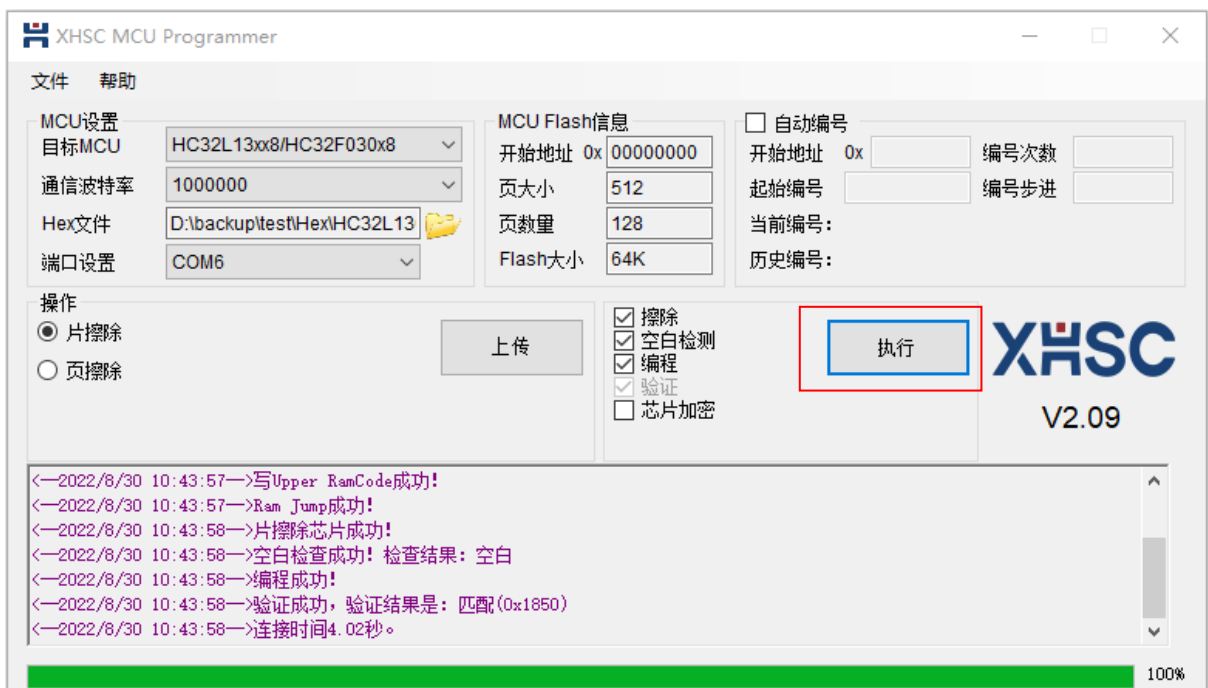


Figure 2-4 Execute Operation

3 Operation Description

3.1 MCU Settings

MCU related information settings, select the MCU model, set the crystal frequency or communication baud rate, select the Hex file to be programmed, and the corresponding port.

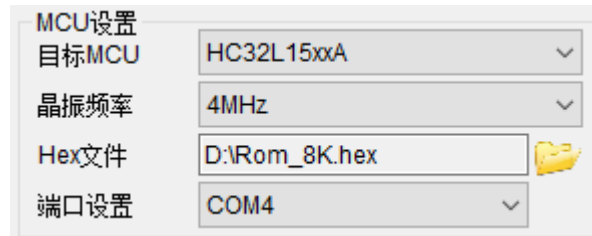


Figure 3-1 MCU Settings

- Target MCU: This drop-down box contains all the ARM cortex-M core MCU types currently available at Xiaohua. Select a model corresponding to the programming target MCU.
- Crystal frequency or communication baud rate: For HC32F146xA/ HC32M140xA and HC32F146x8/
HC32M140x8 series MCUs, this option is to set the crystal frequency used by the target MCU; for other series MCUs, this option is to set the communication baud rate of the selected port.
- Hex file: used to select the Hex file to be programmed.
- Port settings: Set the port number for ISP connection.

3.2 MCU Flash Information

The MCU Flash information of the selected model is displayed: start address, page size, number of pages and Flash size. As shown in Figure 3-2.

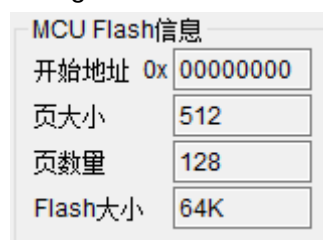


Figure 3-2 MCU Flash information

3.3 Automatic Numbering

本软件带有给目标 MCU 编程时进行编号的功能，只需要在“自动编号”组内选择单选框“自动编号”就可设置自动编号。



☒ 自动编号		
开始地址	0x 2000	编号次数 12
起始编号	1	编号步进 1
当前编号:		
历史编号:		

图 3-3 开启自动编号

选择了“自动编号”后，自动编号的功能使能，填入相应的参数，相关参数描述如下：

- 开始地址：将编号写入到 MCU Flash 地址，起始编号需要占用 4 个字节，地址必须在 MCU Flash 范围内（根据相应型号 MCU 而定），地址为十六进制，从 0-F 之间的字符。
- 编号次数：需要写入多少个编号，编号次数必须大于 0，数值范围是 1 到 999999。
- 起始编号：起始编号是编号的开始编号，从 0 到 4、294、967、295 (0xFFFFFFFF)。
- 编号步进：每个编号之间的步进，数值范围是 0 到 999。
- 当前编号：显示这次编程所写入的编号。
- 历史编号：历史编号是显示上次成功写入的编号。

3.4 Operation

该部分包括本软件的主要功能，上传、擦除、空白检测、编程（验证）和芯片加密。



图 3-4 操作

如果目标 MCU 选择为 HC32F460xExx/ HC32F45xxExx，除上述功能外还会显示：读保护 1、读保护 2 和加密功能。



图 3-5 HC32F460xExx/ HC32F45xxExx 可用的操作

<通用操作>

- **上传：**如果用户有选择烧录的文件，则只读取上传烧录部分内容；
如果用户未选择烧录的文件或者烧录的文件地址不对，则读取上传目标 MCU 整个 Flash 内容。
- **擦除：**擦除包含两种擦除方式：片擦除和页擦除，片擦除是擦除整个芯片，页擦除是根据编程的 Hex 文件所占 Flash 页的位置进行擦除。
- **空白检测：**检查整个芯片是不是属于空白状态（全 0xFF）。
- **编程：**编程是将 Hex 文件编程到 MCU 的操作。
- **验证：**验证编程进去的代码是否正确。

<通用保护功能>（除 HC32F460xExx/ HC32F45xxExx 之外其它芯片的保护功能）

- **芯片加密：**对芯片 flash 数据进行保护，保护后 flash 数据无法被读取。
 - **注：**HC32x07xxAxx/ HC32x17xxAxx/ HC32x19xxCxx 系列芯片加密操作只允许进行 64 次，64 次之后芯片锁死，无法进行解密操作。

<HC32F460xExx/ HC32F45xxExx 芯片保护功能>

- **读保护 1:** 针对 FLASH 区域实行读保护, 以防不受信任的用户读取 FLASH 数据, 保护后通过密钥可以读取 flash 内数据。
- **读保护 2:** 针对 FLASH 区域实行读保护, 以防不受信任的用户读取 FLASH 数据, 保护后任何操作都无法读取 flash 内数据。
- **加密:** 对 FLASH 数据进行加密, 防止 FLASH 遭受物理解析攻击。选中编程选项框的前提下, 选中加密选项框, 并设置加密扇区, 对设置的扇区成功编程后这些扇区的数据被加密。

3.5 Information Display

用于显示操作信息, 以及操作进度。



图 3-6 信息显示

3.6 Command Line Operation

程序支持在 CMD.exe 或者第三方程序中使用命令行操作。以 CMD.exe 为例：

打开 CMD.exe，进入 xhsc.exe 所在文件目录，如图 3-7。



图 3-7 进入程序所在目录

键入“xhsc”，打开 xhsc.exe 程序，进行“MCU 设置”，然后关闭 xhsc.exe 程序。如图 3-8。

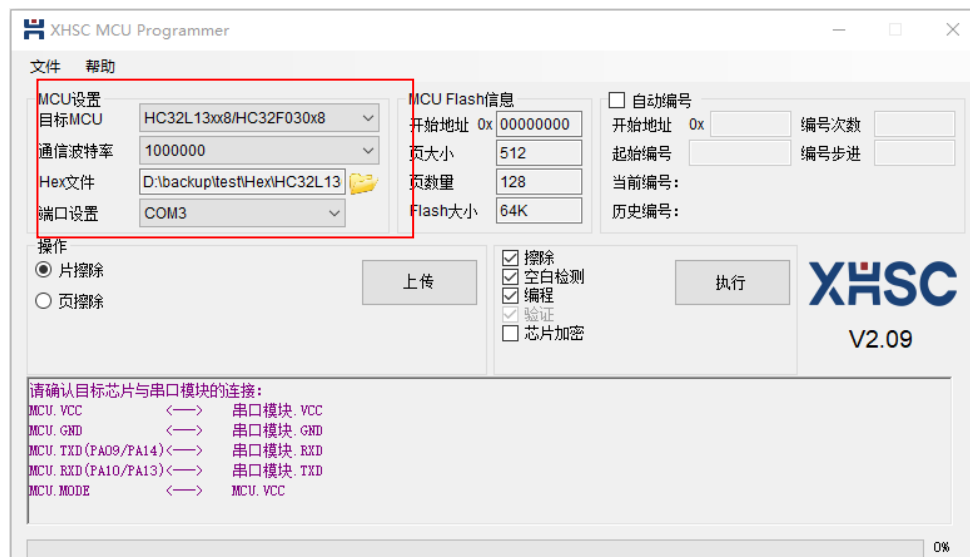


图 3-8 进行 MCU 设置

返回 CMD.exe，键入 xhsc ?，显示可支持的操作命令。如图 3-9。

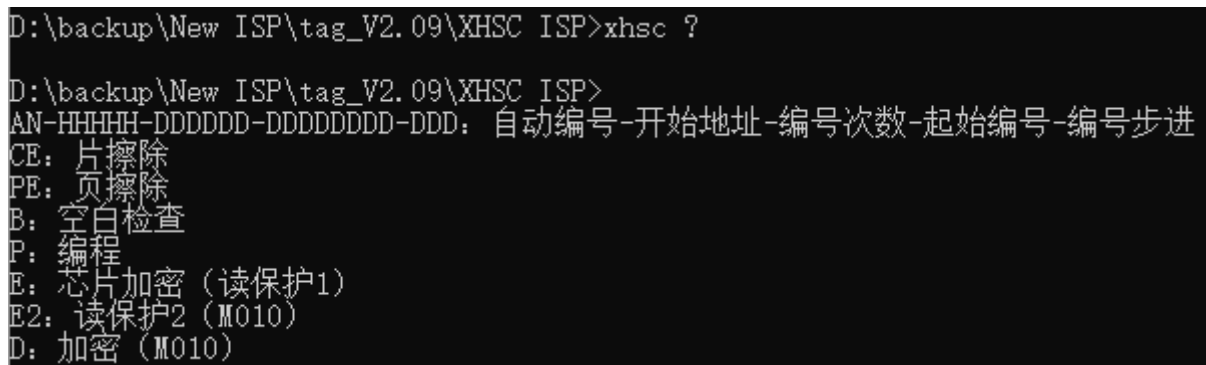


图 3-9 可支持的操作命令

例如使能自动编号，开始地址设置为 0x2000，编号次数设置为 1，起始编号设置为 0，编号步进设置为 1，同时选择片擦除，空白检查和编程，键入“xhsc AN-2000-1-0-1 CE B P”回车，运行结果如图 3-10。

```
D:\backup\New ISP>tag_V2.09\XHSC ISP>xhsc AN-2000-1-0-1 CE B P
D:\backup\New ISP>tag_V2.09\XHSC ISP>连接成功!
下载并运行RAMCODE成功!
更改波特率设置成功!
写Upper RamCode成功!
Ram_Jump成功!
片擦除芯片成功!
空白检查成功! 检查结果: 空白
编程成功!
自动编号: 验证成功, 验证结果是: 匹配(0x0000)
验证成功, 验证结果是: 匹配(0x1850)
连接时间3.14秒。
```

图 3-10 示例操作信息

注意:

- 不同操作命令参数之间需用空格隔开。

4 Error Handling

4.1 Tips

表 4-1 提示

提示信息	描述	处理
连接成功	连接成功	
MCU 已被加密，需要手动重启 MCU，请在手动重启 MCU 后点击“是(Y)”	提示手动重启 MCU	手动重启 MCU, 然后点击“是”按钮继续。
自动编号已完成	自动编号完成	
自动编号地址与用户代码重叠，是否继续操作？	询问是否继续自动编号	点击“是”继续编号，否则停止写入编号

4.2 Errors

表 4-2 错误

错误信息	描述	处理
请选择需要烧录的Hex文件	未选取 Hex 文件	选取 Hex 文件
文件路径不正确或文件无效	选取的 Hex 文件路径不存在或者文件无效或被占用。	重新选取 Hex 文件并确保 Hex 文件有效且不被占用
Hex 文件格式错误	Hex 文件格式错误	确认 Hex 文件是否正确
Hex 文件错误！长度超出所选芯片 Flash 大小。	Hex 文件大小超出所选芯片 Flash 大小	重新选择正确的 Hex 文件或重新选择正确的目标 MCU
请给本计算机安装一个串口	本计算机没有串口	安装一个串口
串口操作超时	通信失败	检查硬件连接, 检查固件是否匹配, 尝试重新上电
读取不成功	读取失败	检查硬件连接, 检查固件是否匹配, 尝试重新上电
MCU Flash 被加密	Flash 被加密, 数据无法读出	被加密芯片, Flash 数据不能读出
片擦除芯片不成功	片擦除失败	检查硬件连接, 检查固件是否匹配, 尝试重新上电
页擦除芯片不成功	页擦除失败	检查硬件连接, 检查固件是否匹配, 尝试重新上电
页擦除芯片不成功: 芯片被加密	被加密芯片不能页擦除	选择片擦除
空白检查不成功	空白检查失败	检查硬件连接, 检查固件是否匹配, 尝试重新上电
检验和归零不成功	检验和归零失败	检查硬件连接, 检查固件是否匹配, 尝试重新上电
编程不成功	编程失败	检查硬件连接, 检查固件是否匹配, 尝试重新上电
验证不成功	验证失败	检查硬件连接, 检查固件是否匹配, 尝试重新上电
参数不能为空, 请填写	参数未填写	填写参数
参数格式错误, 请重新填写	参数格式错误	重新填入参数
地址超过所选芯片Flash大小, 请重新填写	起始地址无效	重新输入地址

Revision History

版本号	修订日期	修订内容
Rev1.0	2017/11/10	Cortex-M 在线编程器用户手册初版发布。
Rev1.1	2019/04/09	增加对软件 V1.4 版本的描述。
Rev1.2	2019/04/15	增加支持芯片型号。
Rev2.0	2019/09/03	对软件 V2.0 版本的支持。
Rev2.01	2021/12/31	型号增加。
Rev2.1	2022/08/30	型号增加及公司 Logo 更新。
Rev2.11	2023/04/13	型号更新。
Rev2.12	2023/07/05	去除芯片具体支持型号描述。